The documentation and process conversion measures necessary to comply with this document shall be completed by 22 May 2020.

INCH-POUND

MIL-PRF-19500/700D
20 January 2020
SUPERSEDING
MIL-PRF-19500/700C
30 January 2014

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, FIELD EFFECT, N-CHANNEL, SILICON,
RADIATION HARDENED (TOTAL DOSE AND SINGLE EVENT EFFECTS)
TYPES 2N7494U5, 2N7495U5, AND 2N7496U5,
QUALITY LEVELS JANTXV AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a N-channel, enhancement-mode, MOSFET, radiation hardened (total dose and single event effects (SEE)), power transistor. These transistors include ratings for avalanche energy maximum rating (E_{AS}) and maximum avalanche current (I_{AR}). Two levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Package outline. The device package outline is a surface mount (LCC-18) package in accordance with figure 1.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	P_T (1) T_C = +25°C	P_T T_A = +25°C	$R_{\theta JC}$ (2)	V_{DS}	V_{DG}	V_{GS}	I_{D1} (3) (4) T_C = +25°C	I_{D2} (3) (4) T_C = +100°C	I_S	I_{DM} (5)	T_J and T_{STG}
	W	W	°C/W	V dc	V dc	V dc	A dc	A dc	A dc	A (pk)	°C
2N7494U5	25	1.67	5.0	30	30	±20	12	8	12	48	−55
2N7495U5				60	60	±20	11.7	7.4	12	46.8	to
2N7496U5				100	100	±20	10	6.5	10	40	+150

(1) Derate linearly by 0.2 W/°C for T_C > +25°C.
(2) See figure 2, thermal impedance curves.
(3) The following formula derives the maximum theoretical I_D limit. I_D is limited by package and device construction to 12 amps.

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \text{ X } (r_{DS(on)}) \text{ at } T_{JM}}}$$

(4) See figure 3, maximum drain current graph.
(5) I_{DM} = 4 X I_{D1}; I_{D1} as calculated in note (3).

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics. Unless otherwise specified at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0 mA dc	$V_{GS(TH)1}$ $V_{DS} \geq V_{GS}$ I_D = 1.0 mA dc		Max I_{DSS1} V_{GS} = 0 V_{DS} = 80 percent of rated V_{DS}	Max $r_{DS(on)}$ (1) V_{GS} = 12 V, $I_D = I_{D2}$		E_{AS}
					T_J = +25°C	T_J = +150°C	
	V dc	V dc Min	V dc Max	µA dc	Ω	Ω	mJ
2N7494U5	30	2.0	4.0	10	0.075	0.158	156
2N7495U5	60	2.0	4.0	10	0.080	0.184	87
2N7496U5	100	2.0	4.0	10	0.110	0.250	60

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example, 6.5 for a list of available PINs, and 6.6 for cross-reference information.

1.5.1 JAN certification mark and quality level. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are "JANTXV" and "JANS".

1.5.2 RHA designator. The RHA levels that are applicable for this specification sheet from lowest to highest for quality levels JANS and JANTXV are as follows: "R", "F", "G", and "H".

1.5.3 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.3.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the devices covered by this specification sheet are as follows: "7494", "7495", and "7496".

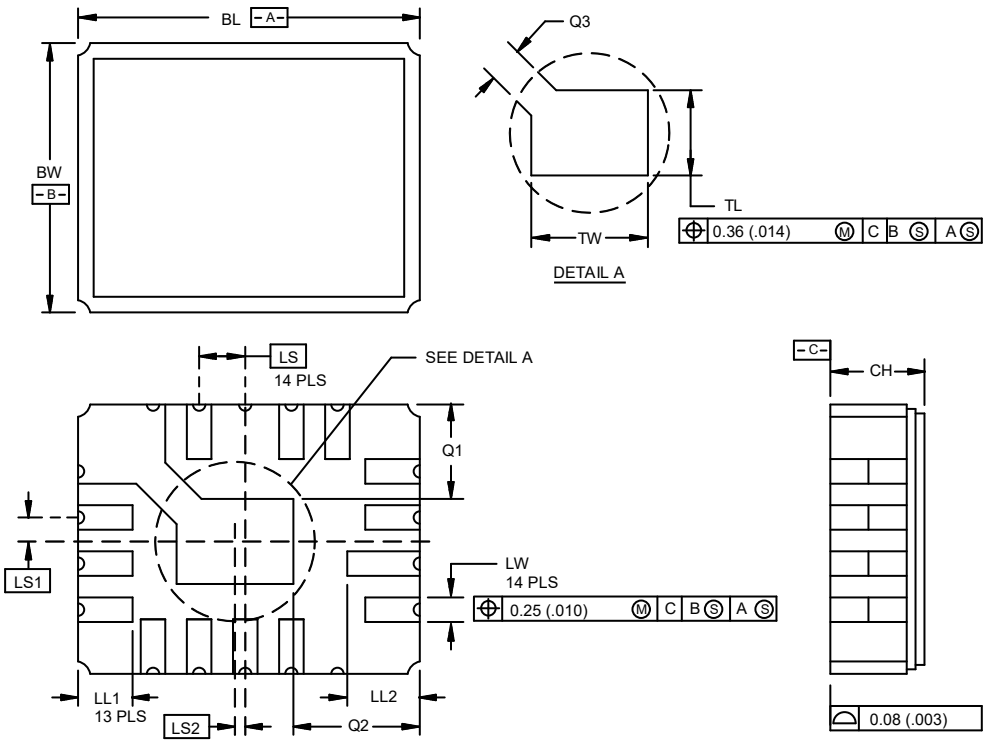
1.5.4 Suffix symbols. The suffix symbols "U5" are incorporated into the PINs for this specification sheet to indicate the package configuration.

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.345	.360	8.76	9.14
BW	.280	.295	7.11	7.49
CH	.095	.115	2.41	2.92
LL1	.040	.055	1.02	1.40
LL2	.055	.065	1.40	1.65
LS	.059 BSC		1.50 BSC	
LS1	.025 BSC		.635 BSC	
LS2	.008 BSC		.203 BSC	
LW	.020	.030	0.51	0.76
Q1	.105 REF		2.67 REF	
Q2	.120 REF		3.05 REF	
Q3	.045	.055	1.14	1.40
TL	.070	.080	1.78	2.03
TW	.120	.130	3.05	3.30

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. The terminal assignments are as follows: D = drain, G = gate, S = source, and NC = not connected.
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for LCC package (2N7494U5, 2N7495U5, and 2N7496U5).

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

I_{AS}	Rated avalanche current, nonrepetitive
nC	nano Coulomb.
SEE	Single Event Effects.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (LCC-18) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction shall not be permitted to meet the requirements of this specification.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor but shall be retained on the initial container.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

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3.8 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, R ≤ or 100 kΩ, whenever bias voltage is applied drain to source.

3.9 RHA. The RHA requirements and test levels shall be as defined in MIL-PRF-19500 and herein.

3.10 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 SEE. SEE shall be performed at initial qualification and after process or design changes which may affect radiation hardness (see tables III and IV). Upon qualification, manufacturers shall provide the verification test conditions from section 5 of method 1080 of MIL-STD-750 that were used to qualify the device for inclusion into section 6 of the slash sheet. End-point electrical measurements shall be in accordance with table II. SEE characterization data shall be made available upon request of the qualifying or acquiring activity.

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4.3 Screening (JANS and JANTXV). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (1) (2)	Measurement	
	JANS	JANTXV
(3)	Gate stress test (see 4.3.1)	Gate stress test (see 4.3.1)
(3)	Single pulse avalanche energy test (see 4.3.2)	Single pulse avalanche energy test (see 4.3.2)
(3) 3c	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
9	Subgroup 2 of table I herein I_{DSS1}, I_{GSSF1}, I_{GSSR1} as a minimum	Not applicable
10	Method 1042 of MIL-STD-750, test condition B	Method 1042 of MIL-STD-750, test condition B
11	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(ON)1}$, $V_{GS(TH)1}$ Subgroup 2 of table I herein. ΔI_{GSSF1} = ±20 nA dc or ±00 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±10 µA dc or ±100 percent of initial value, whichever is greater.	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(ON)1}$, $V_{GS(TH)1}$ Subgroup 2 of table I herein.
12	Method 1042 of MIL-STD-750, test condition A	Method 1042 of MIL-STD-750, test condition A
13	Subgroups 2 and 3 of table I herein ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±10 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(ON)1}$ = ±20 percent of initial value. $\Delta V_{GS(TH)1}$ = ±20 percent of initial value.	Subgroup 2 of table I herein ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±10 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(ON)1}$ = ±20 percent of initial value. $\Delta V_{GS(TH)1}$ = ±20 percent of initial value.

(1) At the end of the test program, I_{GSSF1}, I_{GSSR1}, and I_{DSS1} are measured.
(2) An out-of-family program to characterize I_{GSSF1}, I_{GSSR1}, I_{DSS1}, and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a. Quality level JANTXV does not need to be repeated in screening requirements.

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4.3.1 Gate stress test. Apply V_{GS} = 24 V, minimum for t = 250 µs, minimum.

4.3.2 Single pulse avalanche energy (E_{AS}). The test for single pulse avalanche energy shall be performed in accordance with method 3470 of MIL-STD-750. The following details shall apply:

a. Peak current, $I_{AS} = I_{D1}$.

b. Inductance, $L = (2E_{AS}/(I_{D1})^2)*((V_{BR} - V_{DD})/V_{BR})$ mH minimum.

c. Gate to source resistor, R_{GS}: 25 Ω ≤ R_{GS} ≤ 200 Ω.

d. Supply voltage, V_{DD} = 25 V dc, except V_{DD} = 50 V dc for 2N7496U5.

e. Initial case temperature, T_C = +25 °C, −5 °C, +10 °C.

f. Gate voltage, V_{GS} = 12 V dc.

g. Number of pulses to be applied: 1 pulse minimum.

4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, t_{MD} (and V_H where appropriate). See table III, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JANTXV) of MIL-PRF-19500, and as follows.

4.4.2.1 Quality level JANS.

Subgroup	Method	Condition
B3	1051	Test condition G, 100 cycles
B3	2077	Scanning electron microscope (SEM).
B4	1042	Intermittent operation life, condition D. No heat sink or forced-air cooling on the device shall be permitted during the on cycle. t_{on} = 30 seconds minimum.
B5	1042	Accelerated steady-state gate bias, condition B, V_{GS} = rated; T_A = +175 °C, t = 24 hours minimum; or T_A = +150 °C, t = 48 hours minimum.
B5	1042	Accelerated steady-state reverse bias, condition A, V_{DS} = rated; T_A = +175 °C, t = 120 hours minimum; or T_A = +150 °C, t = 240 hours minimum.
B5	2037	Bond strength, test condition D.

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4.4.2.2 Quality level JANTXV.

Subgroup	Method	Condition
B2	1051	Test condition G, 25 cycles.
B3	1042	Intermittent operation life, condition D. No heat sink or forced-air cooling on the device shall be permitted during the on cycle. t_{on} = 30 seconds minimum.
B5		Not applicable.
B6		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows.

Subgroup	Method	Condition
C2	2036	Terminal strength is not applicable.
C5	3161	Thermal resistance, see 4.3.3, $R_{\theta JC(max)}$ = 5 °C/W.
C6	1042	Intermittent operation life, condition D, 6,000 cycles. No heat sink or forced-air cooling on the device shall be permitted during the on cycle. t_{on} = 30 seconds minimum.

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table E-VIII of MIL-PRF-19500 and table II herein.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein.

4.4.5.1 SEE. Design capability shall be tested on the initial qualification and thereafter whenever a major die design or process change is introduced. See the design safe operation area figure.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. The conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.3.3	$Z_{\theta JC}$			°C/W
Breakdown voltage drain to source	3407	Bias condition C, V_{GS} = 0, I_D = 1 mA dc	$V_{(BR)DSS}$			
2N7494U5				30		V dc
2N7495U5				60		V dc
2N7496U5				100		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)1}$	2.0	4.0	V dc
Gate current	3411	V_{GS} = +20 V dc, bias condition C, V_{DS} = 0	I_{GSSF1}		+100	nA dc
Gate current	3411	V_{GS} = −20 V dc, bias condition C, V_{DS} = 0	I_{GSSR1}		−100	nA dc
Drain current	3413	V_{GS} = 0, bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		10	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 12 V dc, condition A, pulsed (see 4.5.1), I_D = I_{D2}	$r_{DS(ON)1}$			
2N7494U5					0.070	Ω
2N7495U5					0.080	Ω
2N7496U5					0.110	Ω
Forward voltage	4011	V_{GS} = 0, condition A, pulsed (see 4.5.1), I_D = I_{D1}	V_{SD}		1.8	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 3						
High temperature operation		$T_C = T_J = +125$ °C				
Gate current	3411	$V_{GS} = \pm 20$ V dc, bias condition C, $V_{DS} = 0$	I_{GSS2}		±200	nA dc
Drain current	3413	$V_{GS} = 0$, bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS2}		25	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 12 V dc, condition A, pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(ON)3}$			
2N7494U5					0.140	Ω
2N7495U5					0.160	Ω
2N7496U5					0.220	Ω
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)2}$	1.0		V dc
Low temperature operation		$T_C = T_J = -55$ °C				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS(TH)3}$, I_D = 1 mA dc	$V_{GS(TH)3}$		5.0	V dc
Subgroup 4						
Forward transconductance	3475	$I_D = I_{D2}$, V_{DD} = 15 V dc (see 4.5.1)	g_{FS}			
2N7494U5				8.0		S
2N7495U5				7.0		S
2N7496U5				5.0		S
Switching time test	3472	$I_D = I_{D1}$, V_{GS} = 12 V dc, R_G = 7.5 Ω, V_{DD} = 50 percent of rated V_{DS}				
Turn-on delay time			$t_{D(on)}$		25	ns
Rise time			t_r		100	ns
Turn-off delay time			$t_{D(off)}$		35	ns
Fall time			t_f		30	ns

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 5						
Safe operating area test (high voltage)	3474	See figure 4, tp = 10 ms minimum, V_{DS} = 80 percent of maximum rated V_{DS}				
Electrical measurements		See table I, subgroup 2				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B, $I_D = I_{D1}$, V_{GS} = 12 V dc, V_{DD} = 50 percent of rated V_{DS}				
On-state gate charge			$Q_{G(ON)}$			
2N7494U5					65	nC
2N7495U5					45	nC
2N7496U5					50	nC
Gate to source charge			Q_{GS}			
2N7494U5					20	nC
2N7495U5					15	nC
2N7496U5					7.4	nC
Gate to drain charge			Q_{GD}			
2N7494U5					10	nC
2N7495U5					20	nC
2N7496U5					20	nC
Reverse recovery time	3473	di/dt = −100 A/µs, $V_{DD} \leq V_{(BR)DSS} \leq 50$ V, $I_D = I_{D1}$	t_{rr}			
2N7494U5					102	ns
2N7495U5					125	ns
2N7496U5					200	ns

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point electrical measurements, this test is required for the following subgroups:
Group B Subgroups 2 and 3 (JANTXV).
Group B Subgroups 3 and 4 (JANS).
Group C, subgroups 2 and 6.
Group E, subgroup 1.

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TABLE II. Group D inspection.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Pre-irradiation limits		Post-irradiation limits				Unit
				R, F, G, and H		R, F, and G		H 4/		
	Method	Conditions		Min	Max	Min	Max	Min	Max	
Subgroup 1										
Not applicable										
Subgroup 2		T_C = + 25 °C								
Steady-state total dose irradiation (V_{GS} bias) 5/	1019	V_{GS} = 12 V; V_{DS} = 0								
Steady-state total dose irradiation (V_{DS} bias) 5/	1019	V_{GS} = 0; V_{DS} = 80 percent of rated V_{DS} (pre-irradiation)								
End-point electricals										
Breakdown voltage, drain to source	3407	V_{GS} = 0; I_D = 1 mA; bias condition C	$V_{(BR)DSS}$							
2N7494U5				30		30		30		V dc
2N7495U5				60		60		60		V dc
2N7496U5				100		100		100		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = 1 mA	$V_{GS(th)1}$	2.0	4.0	2.0	4.0	1.5	4.0	V dc
Gate current	3411	V_{GS} = +20 V; V_{DS} = 0 bias condition C	I_{GSSF1}		100		100		100	nA dc
Gate current	3411	V_{GS} = −20 V; V_{DS} = 0 bias condition C	I_{GSSR1}		−100		−100		−100	nA dc
Drain current	3413	V_{GS} = 0, V_{DS} = 80 percent of rated V_{DS} (pre-irradiation) bias condition C	I_{DSS}		10		10		25	µA dc
Static drain to source on-state voltage	3405	V_{GS} = 12 V; I_D = I_{D2} condition A pulsed (see 4.5.1)	$V_{DS(on)}$							
2N7494U5					0.192		0.192		0.336	V dc
2N7495U5					0.252		0.252		0.318	V dc
2N7496U5					0.403		0.403		0.504	V dc
Forward voltage source drain diode	4011	V_{GS} = 0 V; I_D = I_{D1} pulsed (see 4.5.1)	V_{SD}		1.8		1.8		1.8	V dc

1/ For sampling plan see MIL-PRF-19500.
2/ Group D qualification may be performed prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other specification utilizing the same die design.
3/ At the manufacturer's option, group D samples need not be subjected to the screening tests, and may be assembled in it's qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.
4/ The H designation represents devices which pass end-point electricals at all 100K, 300K and 600K rads (Si).
5/ Separate samples shall be pulled for each bias.

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TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Temperature cycling	1051	Test condition G, 500 cycles.	
Hermetic seal	1071	Fine and gross leak.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2 1/			45 devices, c = 0
Steady-state gate bias	1042	Condition B, 1,000 hours.	
Electrical measurements		See table I, subgroup 2 herein.	
Steady-state reverse bias	1042	Condition A, 1,000 hours.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			
Not applicable			
Subgroup 10			22 devices, c = 0
Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors	3476	Test conditions shall be derived by the manufacturer.	
Subgroup 11			3 devices
SEE 2/ 3/	1080	See MIL-STD-750 method 1080.	

1/ A separate sample for each test shall be pulled.
2/ Group E qualification of SEE testing may be performed prior to lot formation. Qualification may be extended to other specification sheets utilizing the same structurally identical die design.
3/ Device qualification to a higher level linear energy transfer (LET) is sufficient to qualify all lower level LETs.

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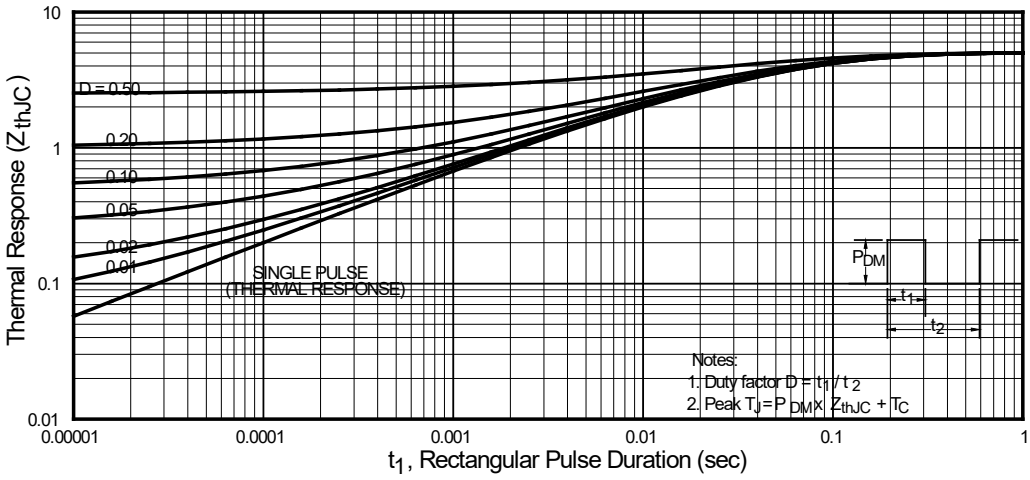
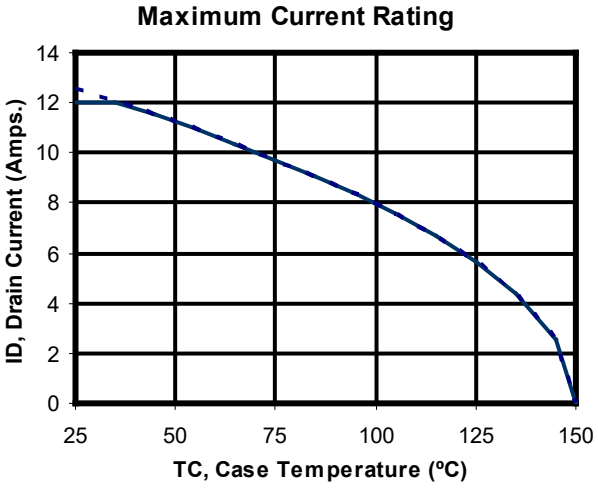
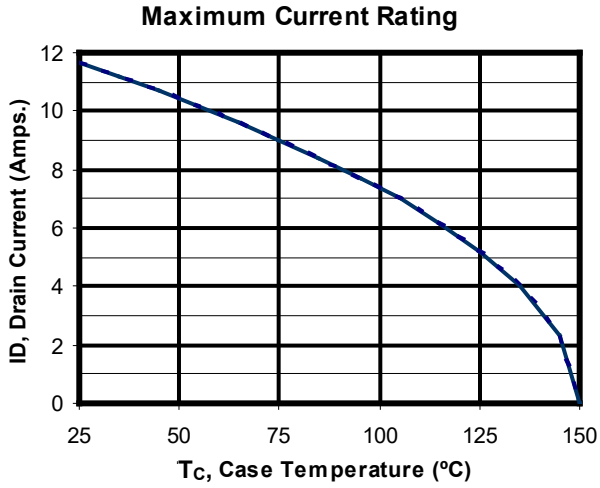


FIGURE 2. Thermal response curve.

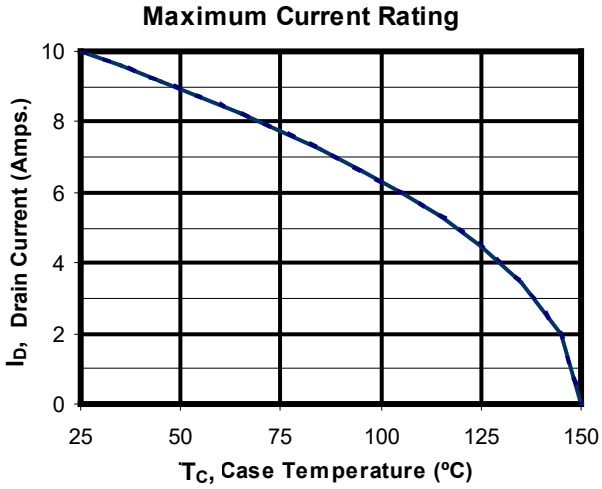
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FIGURE 3. Maximum drain current versus case temperature graphs.

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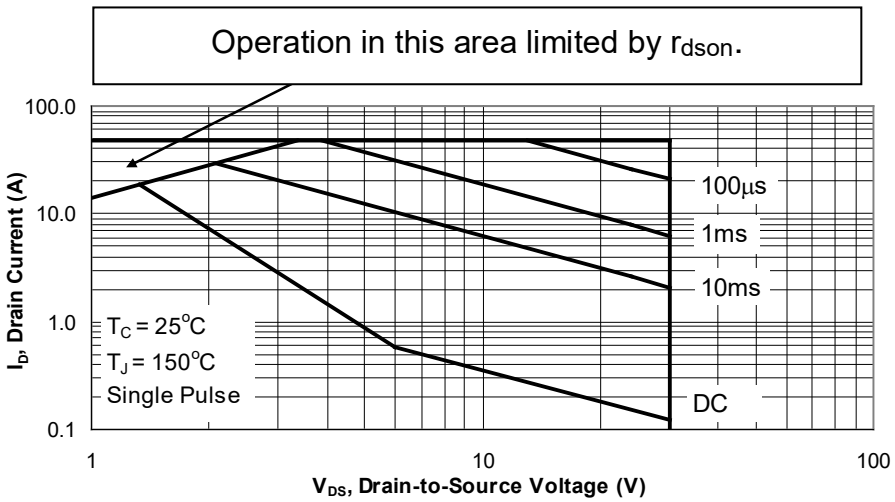


FIGURE 4. Safe operating area graphs.

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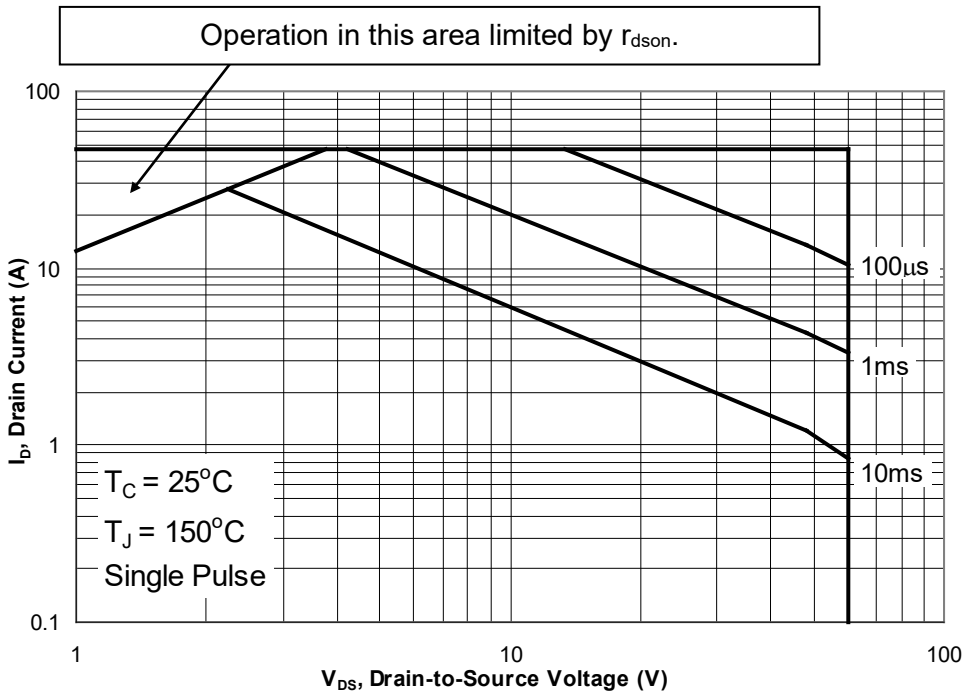


FIGURE 4. Safe operating area graphs - Continued.

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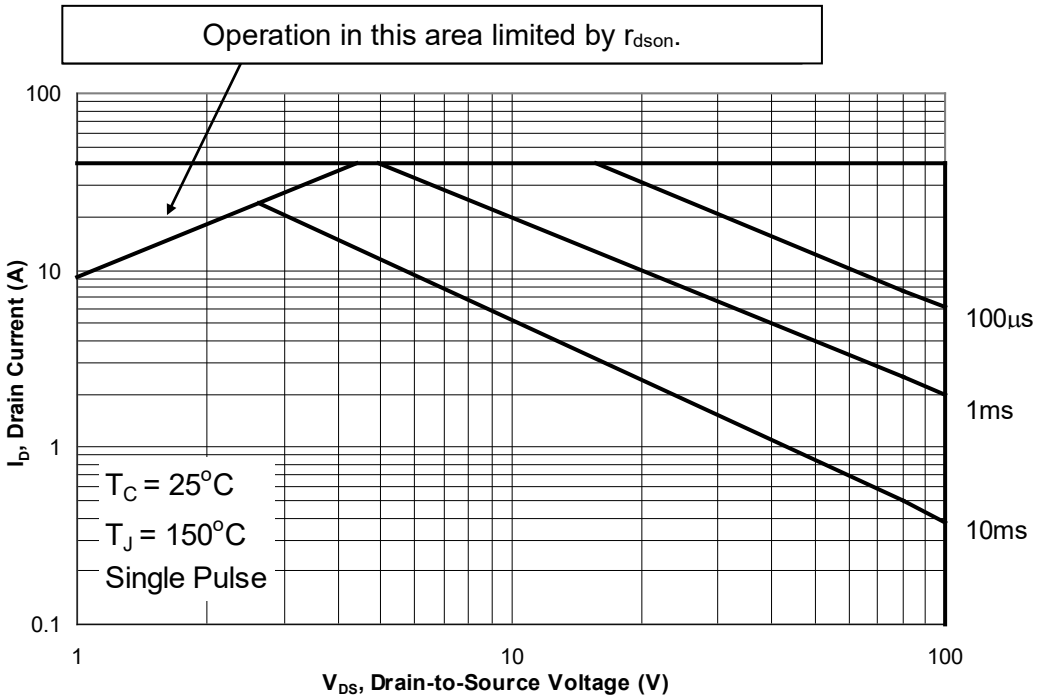


FIGURE 4. Safe operating area graph - Continued.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.4.

e. For acquisition of RHA designated devices, table II, subgroup 1 testing of group D herein is optional. If subgroup 1 is desired, it should be specified in the contract or order.

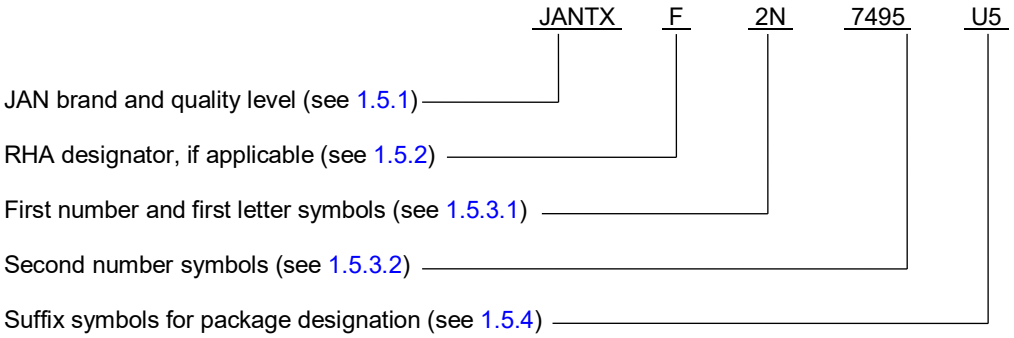
f. If SEE testing data is desired, it should be specified in the contract or order.

g. If specific SEE characterization conditions are desired (see section 6.7 and table IV), manufacturer's cage code should be specified in the contract or order.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

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6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.



6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

PINs for devices of the "F" RHA level	PINs for devices of the "G" RHA level	PINs for devices of the "H" RHA level	PINs for devices of the "R" RHA level
JANTXVF2N7494U5	JANTXVG2N7494U5	JANTXVH2N7494U5	JANTXVRH2N7494U5
JANTXVF2N7495U5	JANTXVG2N7495U5	JANTXVH2N7495U5	JANTXVRH2N7495U5
JANTXVF2N7496U5	JANTXVG2N7496U5	JANTXVH2N7496U5	JANTXVRH2N7496U5
JANSF2N7494U5	JANSG2N7494U5	JANSH2N7494U5	JANSR2N7494U5
JANSF2N7495U5	JANSG2N7495U5	JANSH2N7495U5	JANSR2N7495U5
JANSF2N7496U5	JANSG2N7496U5	JANSH2N7496U5	JANSR2N7496U5

6.6 Cross-reference list. The following table shows the generic P/N and its associated military P/N (without JAN and RHA prefix).

Generic P/N	Military P/N
IRHE57Z30	2N7494U5
IRHE57034	2N7495U5
IRHE57130	2N7496U5

6.7 Application data.

6.7.1 Manufacturer specific irradiation data. Each manufacturer qualified to this slash sheet has characterized its devices to the requirements of MIL-STD-750 method 1080 and as specified herein. Since each manufacturer's characterization conditions can be different and can vary by the version of method 1080 qualified to, the MIL-STD-750 method 1080 revision version date and conditions used by each manufacturer for characterization have been listed here (see table IV) for information only. SEE conditions and figures listed in section 6 are current as of the date of this specification sheet, please contact the manufacturer for the most recent conditions.

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TABLE IV. Manufacturers characterization conditions.

Manufactures cage	Inspection	MIL-STD-750		Sample plan
		Method	Conditions	
69210 (Applicable to devices with a date code of September 2009 and older)	SEE 1/	1080	See MIL-STD-750E method 1080.0 dated 20 November 2006. See figure 5.	
	Electrical measurements		I_{GSSF1}, I_{GSSR1}, and I_{DSS1} in accordance with table I, subgroup 2	3 devices
	SEE irradiation:		Fluence = 3E5 ±20 percent ions/cm^2 Flux = 2E3 to 2E4 ions/cm^2/sec, temperature = 25° ±5°C	
	2N7494U5		Surface LET = 38 MeV-cm^2/mg ±5%, range = 38 µm ±7.5%, energy = 300 MeV ±7.5% In-situ bias conditions: V_{DS} = 30 V and V_{GS} = −10 V V_{DS} = 22.5 V and V_{GS} = −15 V V_{DS} = 15 V and V_{GS} = −20 V (nominal 3.86 MeV/nucleon at Brookhaven National Lab Accelerator)	
	2N7495U5		In-situ bias conditions: V_{DS} = 60 V and V_{GS} = −15 V V_{DS} = 30 V and V_{GS} = −20 V (nominal 3.86 MeV/nucleon at Brookhaven National Lab Accelerator)	
	2N7496U5		In-situ bias conditions: V_{DS} =100 V and V_{GS} = −20 V (nominal 3.86 MeV/nucleon at Brookhaven National Lab Accelerator) Surface LET = 61 MeV-cm^2/mg ±5%, range = 31 µm ±10%, energy = 330 MeV ±7.5%	
	2N7494U5		In-situ bias conditions: V_{DS} = 25 V and V_{GS} = −5 V V_{DS} = 20 V and V_{GS} = −10 V V_{DS} = 15 V and V_{GS} = −5 V V_{DS} = 7.5 V and V_{GS} = −20 V (nominal 2.92 MeV/nucleon at Brookhaven National Lab Accelerator)	
	2N7495U5		In-situ bias conditions: V_{DS} = 46 V and V_{GS} = −5 V V_{DS} = 30 V and V_{GS} = −10 V V_{DS} = 25 V and V_{GS} = −15 V V_{DS} = 15 V and V_{GS} = −20 V (nominal 2.92 MeV/nucleon at Brookhaven National Lab Accelerator)	
	2N7496U5		In-situ bias conditions: V_{DS} = 100 V and V_{GS} = −10 V V_{DS} = 35 V and V_{GS} = −15 V V_{DS} = 25 V and V_{GS} = −20 V (nominal 2.92 MeV/nucleon at Brookhaven National Lab Accelerator)	

See footnotes at end of table.

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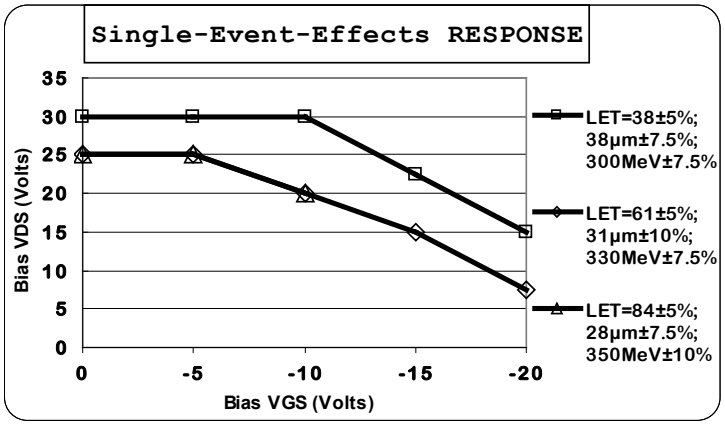
TABLE IV. Manufacturers characterization conditions - Continued.

Manufactures cage	Inspection	MIL-STD-750		Sample plan
		Method	Conditions	
	2N7494U5		Surface LET = 84 MeV-cm^2/mg ±5%, range = 28 µm ±7.5%, energy = 350 MeV ±10% In-situ bias conditions: V_{DS} = 25 V and V_{GS} = −5 V V_{DS} = 20 V and V_{GS} = −10 V (nominal 1.98 MeV/nucleon at Brookhaven National Lab Accelerator)	
	2N7495U5		In-situ bias conditions: V_{DS} = 35 V and V_{GS} = −5 V V_{DS} = 25 V and V_{GS} = −10 V V_{DS} = 15 V and V_{GS} = −15 V V_{DS} = 10 V and V_{GS} = −20 V (nominal 1.98 MeV/nucleon at Brookhaven National Lab Accelerator)	
	2N7496U5		In-situ bias conditions: V_{DS} = 100 V and V_{GS} = −8 V V_{DS} = 80 V and V_{GS} = −10 V V_{DS} = 25 V and V_{GS} = −15 V (nominal 1.98 MeV/nucleon at Brookhaven National Lab Accelerator)	
	Electrical measurements		I_{GSSF1}, I_{GSSR1}, and I_{DSS1} in accordance with table I, subgroup 2	
Upon qualification, all manufacturers should provide the verification test conditions to be added to this table.				

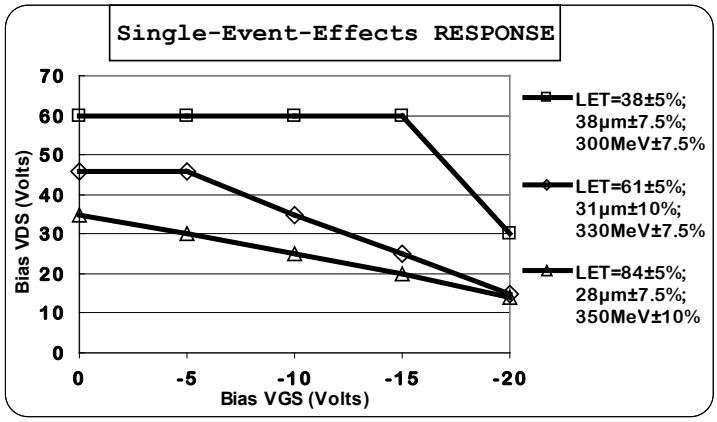
1/ I_{GSSF1}, I_{GSSR1}, and I_{DSS1} was examined before and following SEE irradiation to determine acceptability for each bias condition. Other test conditions in accordance with table I, subgroup 2, may be performed at the manufacturer's option.

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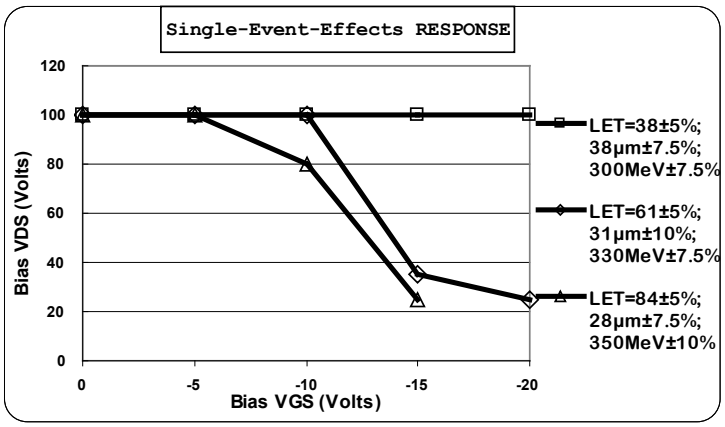


FIGURE 5. Cage 68210 typical SEE response graphs.

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6.8 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

6.9 Changes from previous issue. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army − CR
Navy − EC
Air Force − 85
NASA − NA
DLA − CC

Review activities:
Army − AV, MI
Navy − MC
Air Force − 19, 71

Preparing activity:
DLA − CC

(Project 5961−2019−120)

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